Allen Saji

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SUMMARY

Final-year engineering student at QUT with a hybrid software and electrical background. Project experience spans AWS cloud architecture, React frontend development, embedded C systems, power electronics, and computer vision research. Open to graduate software or electrical engineering roles.

EDUCATION

Queensland University of Technology Expected Dec. 2026
Bachelor of Engineering (Honours), Computer and Software Systems Brisbane, QLD

- Minoring in Advanced Electrical Engineering and Robotics & Embedded Systems
- Dean's List

Indian School Al Ghubra Jun. 2022
High School Diploma – 7.0 GPA Muscat, Oman

- Executive, Robotics Club (2018 – 2020)

EXPERIENCE

Coles May 2024 – Present
Team Member (Service and Online) Brisbane, QLD

- Led day-to-day operations within the online team, coordinating workflows to ensure efficient order fulfilment and high customer satisfaction
- Acted as a first point of contact for customer issues, resolving disputes and escalating when necessary to maintain service quality
- Trained and mentored new team members, improving team performance, consistency, and adherence to procedures
- Managed high-volume transactions using point-of-sale systems with accuracy and efficiency, maintaining smooth store operations

Wildfire Sports & Trek Oct. 2023 – Feb. 2024
Warehouse Dispatcher Brisbane, QLD

- Reduced order processing times with streamlined picking, packing, and shipping procedures
- Participated in continuous improvement initiatives to streamline workflows and optimize productivity
- Prepared shipments for safe transportation by securely wrapping parcels and labeling packages according to company and shipping guidelines

PROJECTS

Portfolio Website | *React 19, Vite, Framer Motion, CSS* Dec. 2025 – Present

- Designed and built allensaji.com end-to-end with no UI framework, developing a custom CSS design-token system that powers synchronized light and dark themes with no flash on page load
- Engineered an interactive project timeline component with distinct desktop and mobile interaction models, click-to-expand detail panels with hover previews on desktop, independent tap-to-toggle cards on mobile that adapt seamlessly as the viewport changes
- Built a shared animation system on top of Framer Motion to drive scroll reveals, staggered transitions, and panel animations consistently across the site, with full support for users who prefer reduced motion
- Developed the entire site to be fully keyboard-navigable and accessible, with responsive layouts tested from mobile through widescreen; deployed on Vercel

Honours Research: Bubble Analysis in Water Electrolysers | *Python, PyTorch, Detectron2* Feb. 2026 – Oct. 2026

- Built a reproducible instance segmentation pipeline for 2,000 fps electrolyser video, auditing three annotators' CVAT exports into a verified dataset of 37,388 bubble labels, split by sequence to prevent data leakage
- Fine-tuned Mask R-CNN in Detectron2 at three input scales and compared it with StarDist and a hybrid using one-shot held-out testing, raising mask AP from 40.4 to 45.1 ($\pm$0.2 over three seeds) with an F1 of 0.80
- Detected 94–100% of bubbles above 25 px, including overlapping ones, and measured gas coverage and Sauter mean diameter to within about 3%; counts ran about 20% low because missed bubbles are numerous but tiny
- Quantified inter-annotator agreement and showed the small-bubble limit below about 15 px lines up with where human labellers stop agreeing, making a consistent labelling rule the top recommendation

Cloud-Based Background Removal System | *AWS, Node.js, Docker* Aug. 2025 – Nov. 2025

- Designed and deployed a cloud-native background removal platform on AWS as two independent containerised services a stateless API tier and an ML processing tier, orchestrated with ECS Fargate for serverless compute and auto-scaling
- Architected an asynchronous SQS pipeline to decouple user requests from ML inference, with retry logic and a Dead Letter Queue to automatically isolate failing jobs after three attempts
- Integrated a neural network-based background removal model into the processor service, with processed images stored in S3 and job status tracked through PostgreSQL on RDS
- Secured the platform with AWS Cognito authentication and S3 pre-signed URLs so image data bypassed the API servers entirely, keeping the compute tier stateless and horizontally scalable; deployed behind an ALB with HTTPS and a custom domain

Solar Inverter Reverse Engineering | *LTspice, MATLAB* Feb. 2025 – Jun. 2025

- Reverse-engineered a commercial 5 kW grid-tied PV inverter with a partner, reconstructing both the DC-DC boost stage and single-phase full-bridge DC-AC stage from physical measurements and datasheet specifications
- Modelled the boost converter in LTspice and MATLAB, validating theoretical ripple and duty cycle calculations against simulation and estimating stage efficiency after accounting for inductor, switching, and capacitor losses
- Designed the full-bridge inverter stage using sinusoidal pulse-width modulation to push switching noise above the audible range, selecting a suitably rated IGBT and sizing the output filter to meet grid voltage and waveform quality requirements
- Proposed a snubber circuit to significantly reduce switching losses in the boost MOSFET, and identified synchronous rectification as a further efficiency improvement path

Elevator System Simulation | *C, POSIX, TCP/IP* Aug. 2024 – Oct. 2024

- Designed and implemented a multi-component elevator system in C for Linux using POSIX shared memory, multithreading, and TCP/IP sockets
- Built modules for car control, scheduling, safety, and user interaction with emphasis on inter-process communication and real-time responsiveness
- Followed safety-critical standards (MISRA C) and developed a floor scheduling algorithm integrated via a custom build system

SKILLS

Programming : Python, C, C#, JavaScript/Node.js, HTML/CSS

Frontend : React, Vite, Framer Motion

Cloud : AWS (ECS Fargate, EC2, SQS, S3, RDS, Cognito, ALB, CloudFront), Docker, Git, Linux (WSL)

Databases : PostgreSQL, REST APIs

Engineering & Simulation : MATLAB/Simulink, LTspice, KiCad/Eagle, AutoCAD

Electrical & Power Systems : DC-DC converters, PV inverter modelling, PCB design & layout, circuit debugging

Embedded Systems : Microcontroller programming (C), real-time systems, UART/SPI/I2C, hardware debugging

Machine Learning & Computer Vision : PyTorch, Detectron2, OpenCV, TensorFlow/Keras, image processing, instance segmentation, neural network integration

Languages : English (Native), Malayalam (Native)